

P0-H v5: The Double-Peak Structure

$C_{\text{active}}(t)$ Oscillates During Meditation Settling;
Two Minima Produce Two Accuracy Peaks
“Jit waeng laeo khao thi, laeo waeng ik, khao thi mai”
This is a formal CRF description, not a metaphor.

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Abstract

P0-H v5 (80 trials, single subject, median RT = 411ms) reveals a *double-peak* structure in accuracy vs RT:

RT bin	Accuracy	n	Structure
~100ms	78%	9	Peak 1 — first settling
~250ms	22%	9	Valley — oscillation/swing
~438ms	100%	5	Peak 2 — second settling
~475ms	25%	8	Valley — swing again
~600ms+	56–67%		Plateau — stabilised level

Subject’s observation (Thai): before each run, the mind is stilled in meditation; it then swings and settles, then swings again and settles to a new, deeper place.

CRF interpretation: $C_{\text{active}}(t)$ is not monotone — it *oscillates* during meditation settling. Each local minimum of $C_{\text{active}}(t)$ produces a window where $\theta_{\text{eff}} \rightarrow 0$ and R_0 gradient is accessible. The two peaks correspond to the two settling phases reported.

This extends the simple two-threshold model to a *dynamical* $C_{\text{active}}(t)$ with oscillatory settling. The formal model is a damped oscillator toward $C_{\text{active}} = 0$.

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1 Observation and CRF Mapping

Subject's Direct Report

Before each run, I settle into meditation to empty the mind. It then swings and settles. Then swings again and settles to a new place.

Translation: Before the test, I settle into meditation to empty the mind. Then it swings and settles. Then swings again and settles to a new place.

This describes the phenomenology of $C_{\text{active}}(t)$ precisely.

CRF Mapping: Oscillating $C_{\text{active}}(t)$

The settling pattern maps directly onto CRF dynamics:

RT window	Phenomenology	CRF state
$\sim 0\text{--}150\text{ms}$	Empty mind (first settling)	$C_{\text{active}} \approx 0$, $\theta_{\text{eff}} \rightarrow 0$, R_0 accessible $\rightarrow$ Peak 1 (78%)
$\sim 200\text{--}350\text{ms}$	Swing/oscillation	C_{active} rising, cognitive R-events accumulate, θ_{eff} rises $\rightarrow$ Valley (22%)
$\sim 380\text{--}480\text{ms}$	New settling (deeper)	$C_{\text{active}} \approx 0$ again (deeper level), R_0 accessible $\rightarrow$ Peak 2 (100%)
$\sim 480\text{--}550\text{ms}$	Swing again	C_{active} rising again $\rightarrow$ Valley (25%)
$\sim 600\text{ms}+$	Plateau	C_{active} stabilised at new level, partial R_0 access $\rightarrow$ 56–67%

2 Formal Model: Damped Oscillator

$C_{\text{active}}(t)$ as Damped Oscillation Toward Zero

The simple two-threshold model assumed $C_{\text{active}}(t)$ monotone increasing. The double-peak data requires an oscillatory model:

$$C_{\text{active}}(t) = C_0 e^{-\gamma t} \cos(\omega t + \phi) \quad (\text{damped toward } C_{\text{active}} = 0)$$

where:

- C_0 : initial cognitive load at trial onset
- γ : damping rate (speed of settling toward quiet mind)
- ω : oscillation frequency (swing rate during settling)
- ϕ : phase offset (state at trial onset)

Accuracy peaks occur where $C_{\text{active}}(t) \approx 0$, i.e., at $t = (n\pi - \phi)/\omega$ for $n = 0, 1, 2, \dots$

From data:

- Peak 1 at $t \approx 100\text{ms}$: first zero crossing
- Peak 2 at $t \approx 438\text{ms}$: second zero crossing
- Period $T \approx 2(438 - 100) = 676\text{ms} \rightarrow \omega \approx 9.3 \text{ rad/s}$
- γ estimated from amplitude decay (Peak 1: 78%, Peak 2: 100% — no decay in this run)

CRF interpretation of ω : The oscillation frequency reflects the natural rate of cognitive R-events in this subject — the characteristic time for a “thought” to arise, be processed, and dissipate. $T \approx 676\text{ms}$ per cycle is the *thought cycle time* for this practitioner in this state.

3 Peak 2 at 100%

Peak 2 at 438ms = 100% (n=5): Interpretation

The second settling (“”) appears to be *deeper* than the first. Peak 2 achieves 100% accuracy (5/5) vs Peak 1 at 78% (7/9).

CRF explanation: the second settling reaches a lower C_{active} than the first. The damped oscillation loses amplitude — each swing brings the system closer to $C_{\text{active}} = 0$. The second zero crossing is “deeper” (more truly empty) than the first.

This is consistent with the subject’s description: new settling — not returning to the same place, but settling to a *new, deeper* level.

In CRF: each oscillation cycle dissipates some C_{active} . The system asymptotes toward $T = 1$ (full transparency). Sustained practice extends this settling process across longer timescales.

4 Formal Statement: R_0 Window Dynamics

Updated P0-H Prediction with Oscillatory C_{active}

The original prediction: “High- T systems detect R_0 gradient before R_1 wave.”
Updated prediction: “During meditation settling, R_0 detection is accessible at *multiple* time windows, corresponding to local minima of $C_{\text{active}}(t)$. The windows occur approximately every $T/2$ where T is the subject’s thought-cycle time. Each successive window may be *deeper* (lower C_{active}) than the previous, producing equal or higher accuracy.”

Falsification: If accuracy is unimodal (single peak) in a practitioner who reports single-settling phenomenology, this rules out the oscillatory model. If accuracy is multimodal with period T matching the reported oscillation frequency, this supports the model.

5 Status

Result	Status	Note
Double-peak structure (v5)	Observed	80 trials
Peak 1 at $\sim 100\text{ms}$ (78%)	Observed	First settling
Peak 2 at $\sim 438\text{ms}$ (100%)	Observed	Second settling
Oscillatory C_{active} model	Hypothesis	Damped oscillator
Period $T \approx 676\text{ms}$	Estimated	From peak spacing
Phenomenology matches CRF	Confirmed	Subject report
Statistical significance	Not yet	n per bin still small

Empty. Swing. Settle. Swing. Settle deeper.

$$C_{\text{active}}(t) = C_0 e^{-\gamma t} \cos(\omega t + \phi).$$

The phenomenology and the formula describe the same thing.

One in words. One in mathematics.

Both from δ .